Dynamic EMS Dispatch & Station Placement Optimization

Station siting and ambulance dispatch, optimised on 2.2 million real Seattle 911 records

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23MNG336 | Operations Research

Two decisions control how fast an ambulance arrives

Q1. Where should ambulances wait?

Place stations well and more of the city sits inside a short response radius. A strategic decision, made once.

Set covering and maximal covering, ILP

Q2. Who should be sent?

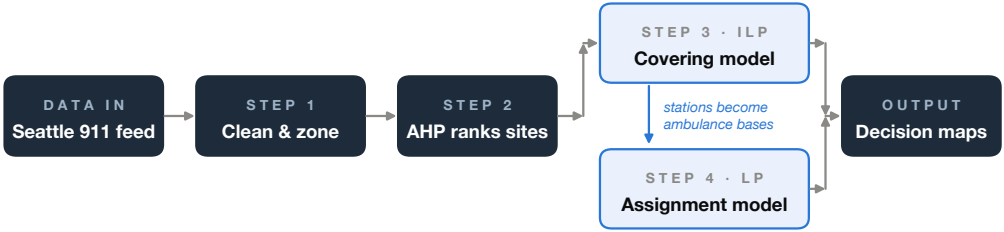
When several emergencies land at once, the right ambulance-to-incident matching cuts everyone's wait. An operational decision, made every hour.

Assignment problem, LP

The two are linked: the stations chosen by the covering model become the ambulance bases that the assignment model dispatches from.

The pipeline

Real data in, four models, decisions out



Every stage runs on the real feed. No synthetic demand, no sampled subset, no hand-picked study area.

All three optimisations solve locally in milliseconds, so the whole pipeline is interactive rather than a batch job.

The data

Three public sources, no API keys, fully reproducible

Source	What it is	Used for
Seattle Real-Time Fire 911 Calls	2,197,739 records, 2003 to 2026	Where and when emergencies occur
Seattle Fire Stations (City GIS)	36 SFD facilities	Benchmarking our placement
ZIP Code Tabulation Areas (Census)	ZIP polygons	Assigning each incident a zone

79,152 EMS incidents
study year, 27 Aug 2025 to 27 Aug 2026

28 demand zones
after a materiality floor

36 real stations
the benchmark network

The raw feed mixes fire, alarms and hazmat with EMS. Calls are filtered to genuine aid and medic dispatches by whole-word patterns, de-duplicated on incident number, and tagged by point-in-polygon join. Of the 36 benchmark facilities, 34 are numbered fire stations; the other two are department headquarters and Harborview Medical Center.

What changed since Review 1

At the first review we presented on the 911 call feed alone, and said we would bring further data in. We did: both of the supporting datasets below were added afterwards, and each one unlocks a part of the project that could not exist without it.

Seattle Fire Stations

36 facilities, City GIS

- Supplies the **Equity Gap** criterion: the distance from each candidate zone to the nearest station Seattle already runs. Without it the AHP has two criteria, not three.
- It *is* the benchmark. Every claim about needing fewer sites than the real network is measured against this file.

ZIP Code Tabulation Areas

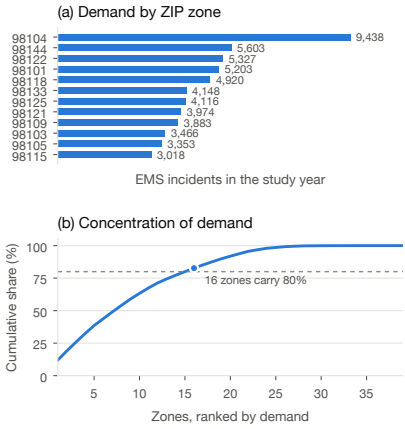
56 polygons, US Census TIGER

- Turns 79,152 scattered points into **28 demand zones** by point-in-polygon join.
- A covering ILP needs discrete demand points and candidate sites. Without the zones there is no tractable model to solve, only a cloud of coordinates.

Both are public, open-licence and fetched by `scripts/download_data.py`, so the three-source pipeline reproduces from a clean checkout.

Step 1. Demand is concentrated, not uniform

Which is why placement can beat history



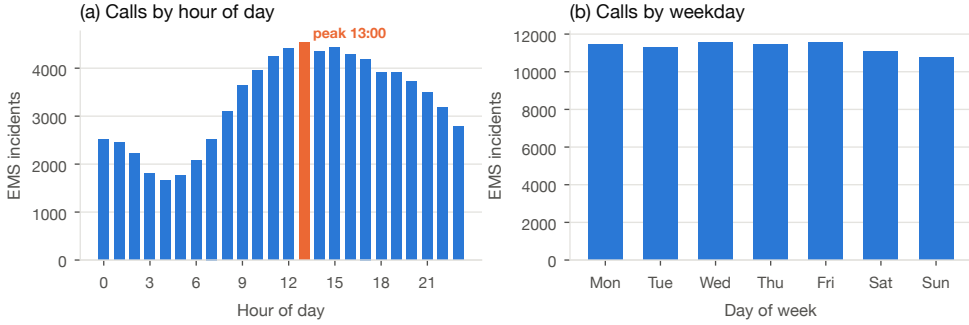
The 15 busiest ZIP zones carry **79.97%** of all 79,152 incidents, and 16 zones take it past four fifths.

Zone 98104 alone accounts for **9,438** calls, which is 11.9% of the city's EMS demand in one ZIP.

Each zone becomes a demand point weighted by its incident count, located at the *mean incident coordinate* rather than the polygon centroid, so the candidate site sits where the calls actually are.

Step 1. When the load spikes

A midday peak, and a flat week



Demand peaks at **13:00** and runs roughly **216 calls per day**. The weekday spread is only a few percent (Friday 11,577 against Sunday 10,757), so the hour matters for dispatch while the day does not. That is why the dispatch scenario is chosen by peak hour rather than by weekday.

Step 2. AHP turns three competing criteria into one score

Criterion	Weight
Incident Density	**67.4%**
Equity Gap	23.1%
Coverage Potential	9.4%

Four team members rate the three criteria pairwise on Saaty's 1 to 9 scale, independently. The matrices are combined by **element-wise geometric mean**, the only average that keeps the matrix reciprocal.

Weights are the principal eigenvector. Pairwise human judgements can quietly contradict each other, so AHP measures that:

$$\mathrm{CI} = \frac{\lambda_{\max} - n}{n - 1}, \qquad \mathrm{CR} = \frac{\mathrm{CI}}{\mathrm{RI}}$$

$\lambda_{\max}$	CI	RI	CR
3.0000	0.0000	0.58	**0.0000**

$\mathrm{CR} < 0.10$ for all four members individually and for the aggregate.

CR is exactly zero by construction rather than by luck: each member's density-against-coverage rating is the product of their other two, so the judgements contain no contradiction for the check to find.

Step 3. Optimisation 1, where to put the stations

Mode A, set covering

Fewest stations that put every zone in reach.

$$\begin{aligned} \min\ & \textstyle\sum_i y_i \\ \text{s.t. } & \textstyle\sum_{i \in N(j)} y_i \ \geq\ 1 \quad \forall j \\ & y_i \in \{0,1\} \end{aligned}$$

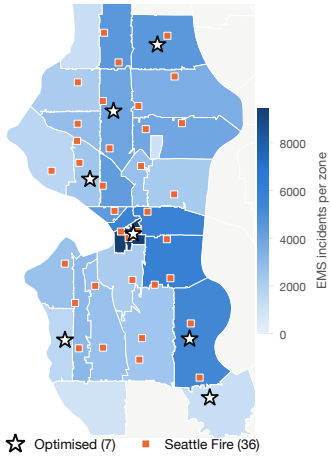
Mode B, maximal covering

On a fixed budget P, cover the most demand.

$$\begin{aligned} \max\ & \textstyle\sum_j d_j z_j \\ \text{s.t. } & z_j \ \leq\ \textstyle\sum_{i \in N(j)} y_i \quad \forall j \\ & \textstyle\sum_i y_i = P, \quad y_i, z_j \in \{0,1\} \end{aligned}$$

$N(j)$ is the set of candidate sites that reach zone j inside the limit. d_j is the zone's incident count, so Mode B maximises people reached rather than area or zone count. Both are solved exactly with HiGHS via PuLP, in **5 ms** on the study instance.

Step 3. The eight-minute solution



7 stations

100.0% of demand covered

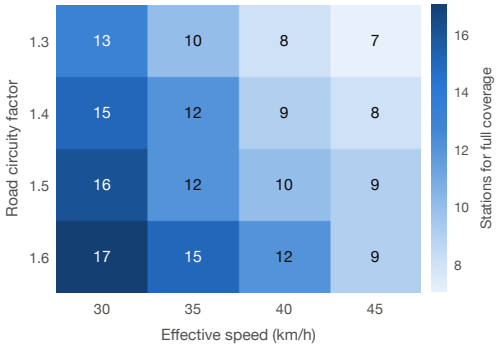
28/28 zones inside 8 minutes

5 ms to prove optimality

Against Seattle's **36** real stations at the same limit.

The next slide shows why 7 should not be quoted on its own.

How much of this is Seattle, and how much is our assumption?



The 911 feed records **no response times**. Every minute in this project is computed:

$$t = \frac{d_{\text{haversine}} \times \text{circuity}}{\text{speed}} + \text{turnout}$$

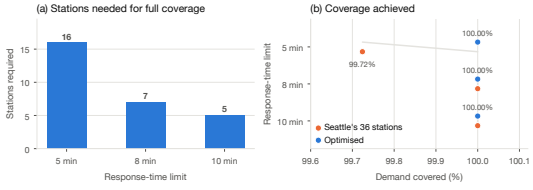
So we re-solved the ILP across a 4×4 grid of those two constants, 16 exact solves.

7 to 17 stations, a 2.4× spread

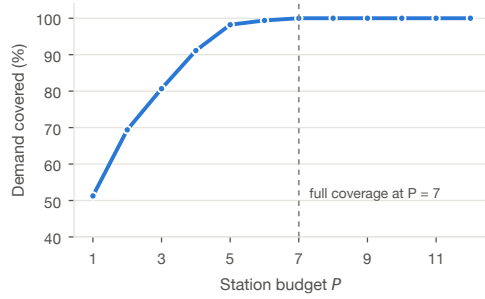
100% coverage in all 16 cells

The count is assumption-dependent. Being under 36 is not.

Step 3. What a tighter promise costs



Stations needed per response limit



Diminishing returns against budget P

Tightening the promise from 10 minutes to 5 takes the network from **5** stations to **16**, so the cost of a guarantee is convex. The budget curve is the argument for Mode B: the first handful of stations buy most of the city, and the last few buy very little.

Step 3. Against the network Seattle actually runs

Response limit	Our stations	Demand covered	Seattle's 36
5 min	16	100%	99.72%
8 min	7	100%	100%
10 min	5	100%	100%

What this supports

Placement that follows measured demand reaches the whole city from fewer sites than placement accumulated historically. Directionally true in every cell of the robustness grid.

What it does not

The 36 are *fire* stations. They also carry suppression, hazmat and technical rescue, missions this model does not place for, so the comparison is not like-for-like and the ratio flatters us.

Step 4. Optimisation 2, who to send

$$
\begin{aligned}
\min \quad & \textstyle\sum_i \sum_j c_{ij}\, x_{ij} \\
\text{s.t.} \quad & \textstyle\sum_i x_{ij} = 1 \quad \forall j \\
& \textstyle\sum_j x_{ij} \le 1 \quad \forall i \\
& x_{ij} \ge 0
\end{aligned}
$$

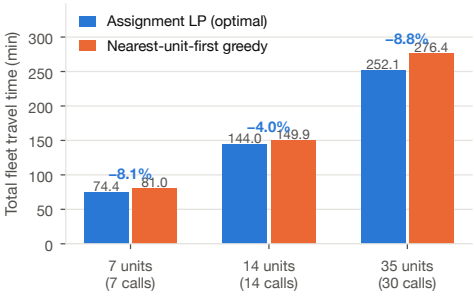
c_{ij} is travel time from ambulance i to incident j. The constraint matrix is totally unimodular, so the LP relaxation is already integral and no branching is needed.

Scenario: the busiest hour in the study year, 09:00 on Wednesday 19 November 2025, with **30 simultaneous incidents**, dispatched from the **7 stations the covering model opened**.

A rectangular cost matrix is padded with zero-cost dummy rows or columns so the problem is square and balanced. Dummy cells cost nothing, so they never distort the real total.

Cross-checked against an independent Hungarian-algorithm solution, which agrees to six decimal places.

Step 4. Optimal dispatch against nearest-unit-first



Fleet	Optimal	Greedy	Gain
35 units *(default)*	252.1	276.4	8.8%
14 units	144.0	149.9	4.0%
7 units	74.4	81.0	8.1%

Both sides clear the same calls. A peak hour has more calls than units, and unserved calls cost nothing in the objective, so an unrestricted LP would quietly answer only the nearest calls while greedy took them as they came.

Comparing those totals would score triage rather than routing, and would report 61.8% instead of 8.1% at seven units.

What is measured, and what is modelled

Measured

- Incident locations, times and counts, from 2.2M actual records
- The 36 real SFD facility locations
- ZIP zone geometry
- Every optimisation result, solved exactly

Modelled

- **Every response time**, since the feed contains none
- Demand aggregated to 28 zone centroids, worth **+1.9 min** on the average call
- AHP weights, subjective by design, hence the CR test
- Units assumed always free, with no queueing

Charging that within-zone travel takes the eight-minute plan from **7 stations to 12**, and the robustness grid cannot see it: scaling speed and circuity moves every distance together. This is an optimisation over **real demand geography** with a **modelled travel layer**. Its claims are therefore comparative, about which sites, in what order, and how far the answer moves, rather than predictive. That is why the robustness grid is a result slide and not an appendix.

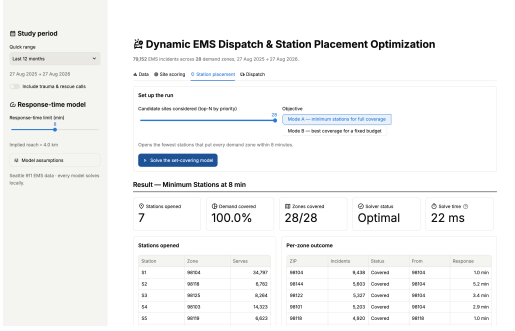
How the models were checked

Each optimisation is validated against an independent method rather than assumed correct:

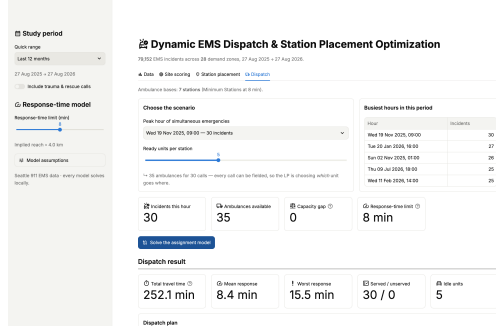
- **Set covering and maximal covering** were re-solved by **brute-force enumeration** on small instances, where every subset of candidate sites can be checked exhaustively. The ILP matched the enumerated optimum on each one.
- **AHP weights** were compared against the largest-modulus eigenvector computed directly by NumPy, over randomly generated reciprocal matrices.
- **The assignment LP** was cross-checked against an independent Hungarian-algorithm implementation, agreeing to six decimal places, and was confirmed never to lose to the greedy baseline over randomly generated cost matrices.
- **The robustness grid** is checked for monotonicity in both directions: a slower fleet or a twistier road network can never need fewer stations. That is what separates a real gradient from solver noise.

The UI implementation

Both optimisations, solved and read in the browser



Station placement: set up, solve, result



Dispatch: the LP on a chosen peak hour

Built with Streamlit. The covering ILP above solves in **22 ms** and its seven stations become the ambulance bases the dispatch LP works from, so the two models are linked in the interface exactly as they are in the formulation.

The application

Four tabs, every model re-solving interactively:

- **Data**, demand in space and time
- **Site scoring**, AHP with live Saaty sliders
- **Station placement**, both ILP modes, sensitivity, robustness and the benchmark
- **Dispatch**, the LP on any peak hour

Every assumption is a control, not a constant:

- Study period and trauma inclusion
- Response-time limit
- Speed, circuity and turnout
- Station budget and fleet size

Built with Streamlit. Every solve runs in-process on the machine serving the app; the only outbound requests are for basemap tiles.

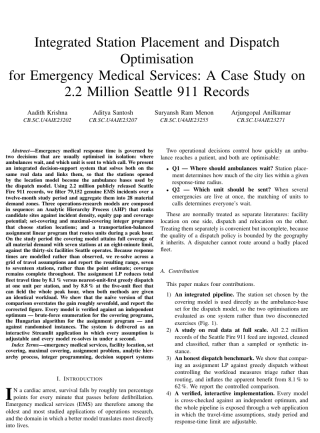
Conclusions

- Demand-driven placement reaches the whole city from **fewer sites** than Seattle's historically accumulated network. The finding is directional and survives every travel assumption we tested.
- The station count itself is **7 to 17**, not 7. Reporting the range is the result; reporting the point estimate would have been the mistake.
- Optimal dispatch beats nearest-unit-first by **8.8%** of total fleet travel time on the busiest hour at the default fleet, measured like-for-like. The margin depends on fleet size, so it is only meaningful quoted with its configuration.
- The two models **compose**: covering output becomes assignment input, so a siting decision is evaluated by the dispatch performance it enables.

Next: replace the modelled travel layer with real road-network distances, and extend the covering model to **MEXCLP** (Daskin, 1983) so that unit busy-fraction is priced in rather than assumed away.

The report

Written to the IEEE conference format



Integrated Station Placement and Dispatch Optimisation for Emergency Medical Services: A Case Study on 2.2 Million Seattle 911 Records

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Abstract—Emergency medical response time is governed by two decisions that are usually optimised in isolation: where ambulances wait, and which unit is sent to which call. We present an integrated decision-support system that solves both on the same real data and links them, so that the stations opened by the placement model are exactly the ones the dispatch and the dispatch model. Using 2.2 million publicly released Seattle Fire 911 records, we filter 79,102 genuine EMS incidents over a twelve-month study period and aggregate them into 28 material demand zones. Three operations-research models are composed in sequence: an Analytic Hierarchy Process (AHP) that ranks candidate sites against incident density, equity gap and coverage redundancy; a bi-objective facility location model (NSGA-II) that chose station locations; and a transportation-balanced assignment linear program that routes units during a peak hour. On the study period the covering model attains full coverage of all material demand with seven stations at an eight-minute limit, while the median sits at four minutes and twenty seconds. Stations are modelled rather than observed, so results across a grid of travel assumptions and report the resulting range, seven to seventeen stations, rather than the point estimate; coverage remains complete throughout. The assignment LP reduces total fleet travel time by 8.1 % versus nearest-unit-first greedy dispatch at one unit per station, and by 8.9 % at the five-unit fleet that the city runs, while cutting the busiest unit's workload under an identical workload. We show that the naive version of that comparison overstates the gain roughly sevenfold, and report the corrected figure. Every model is verified against an independent optimum — brute-force enumeration for the covering program, the Hungarian algorithm for the assignment program — and ships with a reproducible pipeline and a command-line and an interactive Streamlit application in which every assumption is adjustable and every model re-solves in under a second.

Index Terms—emergency medical services, facility location, set covering, maximal covering, assignment problem, analytic hierarchy process, integer programming, decision support systems

I. Introduction

In a cardiac arrest, survival falls by roughly ten percentage points for every minute that passes before defibrillation. Emergency medical services (EMS) are therefore among the oldest and most studied applications of operations research, and the domain in which a better model translates most directly into lives.

Two operational decisions control how quickly an ambulance reaches a patient, and both are optimisable:

- **Q1 — Where should ambulances wait?** Station placement determines how much of the city lies within a given response-time radius.
- **Q2 — Which unit should be sent?** When several emergencies are live at once, the matching of units to calls determines everyone's wait.

These are normally treated as separate literatures: facility location on one side, dispatch and relocation on the other. Treating them separately is convenient but incomplete, because the quality of a dispatch policy is bounded by the geography it inherits. A dispatcher cannot route around a badly placed fleet.

A. Contribution

This paper makes four contributions.

1) **An integrated pipeline.** The station set chosen by the covering model is used directly as the ambulance-base set for the dispatch model, so the two optimisations are evaluated as one system rather than two disconnected exercises (Fig. 1).
2) **A study on real data at full scale.** All 2.2 million records of the Seattle Fire 911 feed are ingested, cleaned and classified, rather than a sampled or synthetic instance.
3) **An honest dispatch benchmark.** We show that comparing an assignment LP against greedy dispatch without controlling the workload measures triage rather than routing, and inflates the apparent benefit from 8.1 % to 62 %. We report the controlled comparison.
4) **A verified, interactive implementation.** Every model is cross-checked against an independent optimum, and the whole pipeline is exposed through a web application in which the travel-time assumptions, study period and response-time limit are adjustable.

Ten pages in `IEEEtran`, two-column conference style, carrying the full treatment:

- Related work across covering location, dispatch and AHP
- Every model stated formally, with the travel-time derivation
- Results, the robustness grid and the study-window sensitivity
- A limitations section that quantifies what the model does not capture
- Eight vector figures, drawn from the same models and the same colour tokens as this deck and the application

Thank you

Questions

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